Global Sequence Alignment

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Global alignment vs Local alignment

1. Global alignment: In the single pair-wise version of the problem we have two DNA, RNA, or amino acid strings *u* and *v* and want to align them completely with one another. Calculating a global alignment is a form of global optimization that forces the alignment to span the entire length of all query sequences. In the multiple version of the problem we have multiple DNA, RNA, or amino acid strings *u1* through *un* and want to align them completely with one another. The latter approaches often use pair-wise alignments as an ingredient.

2. Local alignment/ Alignment search: By contrast to global alignments, local alignments identify regions of similarity within long sequences that are often widely divergent overall. Local alignments are often preferable, but can be more difficult to calculate because of the additional challenge of identifying the regions of similarity.

```
Global FTFTALILLAVAV
       F--TAL-LLA-AV

Local  FTFTALILL-AVAV
       --FTAL-LLAAV--
```

Sequence Alignment

- **Sequence alignment** is a way of arranging two or more sequences of characters to *identify regions of similarity*
  - b/c similarities may be a consequence of functional or evolutionary relationships between these sequences.

- Another definition: Procedure for comparing two or more sequences by searching for a series of individual characters that are *in the same order* in those sequences
  - **Pair-wise alignment**: compare two sequences
  - **Multiple sequence alignment**: compare > 2 sequences

Sequence alignment: example

```
abcdef
| |
abdgf
```

- In order to *maximise the alignment*, we insert gap between **b** and **d** in lower sequence to allow **d** and **f** to align

```
abcdef
| |  |  |
ab-dgf
```

- Note **e** and **g** don't match

Scoring schemes: PAM and BLOSUM

- Substitution scores of both PAM and BLOSUM matrices are derived from the analysis of known alignments of evolutionary related proteins.

- Point Accepted Mutation (PAM), is the point mutation per 100 amino acids:
  - PAM1 means a 1 point mutation/100 amino acids.
  - Different PAM matrices are derived from the multiplication of the PAM1 matrix; as multiple substitutions can occur at the same site

- The BLOSUM matrices are newer and considered better.
  - E.g., BLOSUM62 is the matrix calculated by using the observed substitutions between proteins which have at most 62% sequence identity, etc.

BLOSUM62

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4

Gap penalties

- *Constant gap penalty.* Constant gap penalty means that any gap, whatever size it is, receives the constant negative penalty, $-g$.
  - The total number of gaps matters not their length.
  - *Minimizes the number of gaps.*

- *Linear gap penalty.* Linear gap penalty depends linearly on the size of a gap. Parameter, $-g$, is the <u>penalty per unit</u> length of a gap.
  - The overall penalty for one large gap is the same as for many small gaps.

Needleman-Wunsch algorithm

- We have two 2D matrices: the *score matrix* and the *traceback matrix*.

- The Needleman-Wunsch algorithm consists of 3 steps:
  - Initialisation of the score and the traceback matrices
  - Calculation of scores and filling in the score and traceback matrices
  - Inferring the alignment from the traceback matrix

- In the example we align 2 sequences of amino acids SEND and AND with the BLOSUM62 substitution matrix and the constant gap penalty **g = –10.**

Step 1: initialisation

- During the initialisation, the 1st row and 1st columns of the score and traceback matrices, **Score** and **Tr**, respectively, are initialised in the following way:

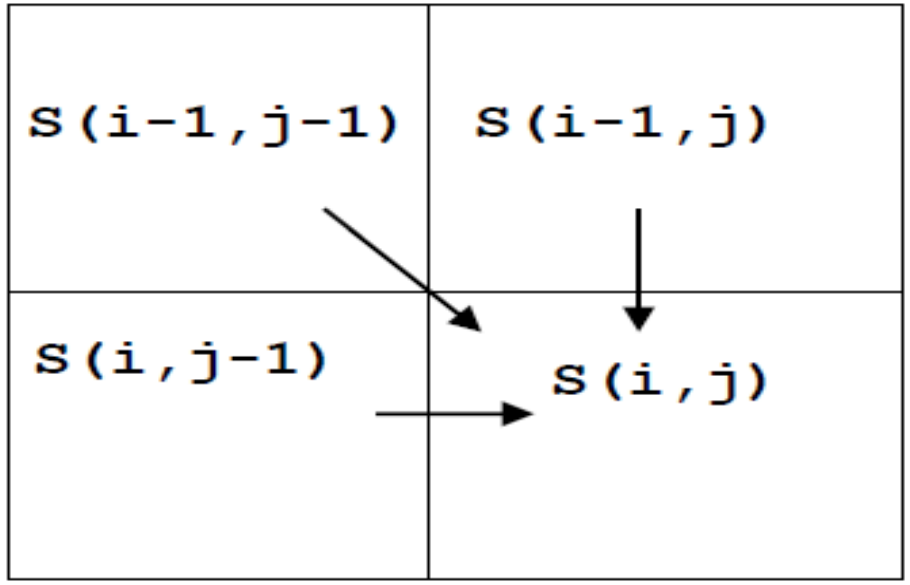
Score		S	E	N	D
	0	-10	-20	-30	-40
A	-10	?			
N	-20				
D	-30				

Tr		S	E	N	D
	done	left	left	left	left
A	up	?			
N	up				
D	up				

The next step is to determine S(2, 2) and T(2, 2)

Step 2: calculation of scores

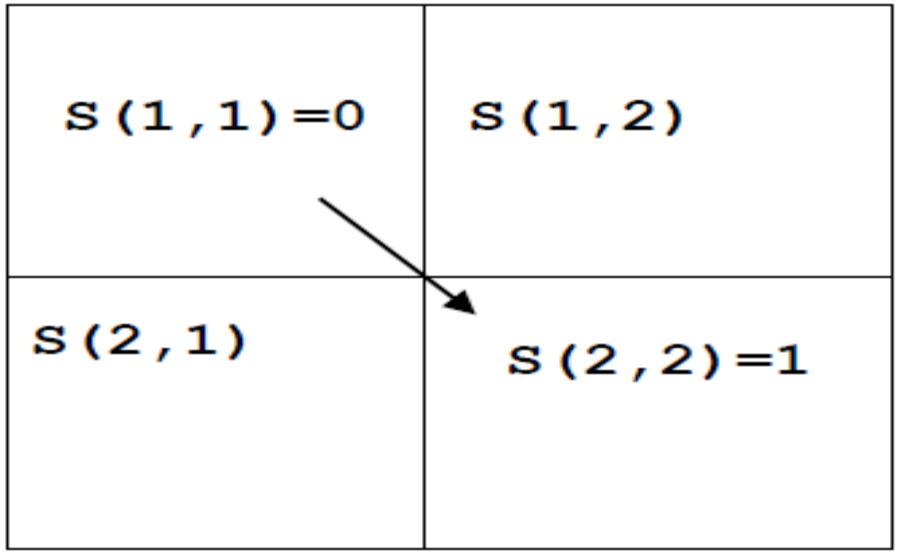
- The next step is to find the score value for an element S(2, 2).
- Rule: Value S(i, j) will become the **maximum** of:
  - diag = S(i – 1, j – 1) + r(i, j) // r is char replacement score
  - up = S(i – 1, j) + g
  - left = S(i, j – 1) + g



Step 2: calculation of scores

- The substitution score for letters i and j according to the BLOSUM62 is denoted as **r(i, j)** and g = –10 is the gap penalty.

- Thus for S(2, 2) we have:
  - diag = S(1, 1) + r(A, S) = 0 + 1 = 1 // this is maximum score
  - up = S(1, 2) + g = –20
  - left = S(2, 1) + g = –20



Step 2: filling in the score matrix

- We calculate the matrix S elements iteratively. Resulting matrix looks like this:

Score		S	E	N	D
	0	-10	-20	-30	-40
A	-10	1	-9	-19	-29
N	-20	-9	-1	-3	-13
D	-30	-19	-11	2	3

Step 2: filling in the traceback matrix

- The traceback items are indices of maximal scores. Both matrices look like this:

Score		S	E	N	D
	0	-10	-20	-30	-40
A	-10	1	-9	-19	-29
N	-20	-9	-1	-3	-13
D	-30	-19	-11	2	3

Tr		S	E	N	D
	done	left	left	left	left
A	up	diag	left	left	left
N	up	diag	diag	diag	left
D	up	up	diag	diag	diag

Step 3: deducing the best alignment

- Traceback is the process of deduction of the best alignment from the traceback matrix.
- The traceback always begins with the last cell, i.e. the bottom rightmost cell in **Tr**.
- Sequences are aligned *backwards*, i.e. from right to left.

```
Resulting alignment:

1.        D
          D

2.       ND
         ND

3.      END
        -ND

4.     SEND
       A-ND
```

Step 3: deducing the best alignment

- There are 3 possible moves along the *traceback path*:
  - Diag: the letters from two sequences are aligned
  - Left: gap is introduced into the left sequence
  - Up: a gap is introduced into the top sequence

Tr		S	E	N	D
	done	left	left	left	left
A	up	diag	left	left	left
N	up	diag	diag	diag	left
D	up	up	diag	diag	diag

Resulting alignment:

```
1.        D
          D

2.       ND
         ND

3.      END
        -ND

4.     SEND
       A-ND
```

Evaluation of the alignment

- Let us evaluate, i.e.score, all possible alignments :

```
SEND
-AND   score = +1
A-ND   score = +3 ← the best
AN-D   score = -3
AND-   score = -8
```

- Thus, the global alignment found by the NW algorithm is indeed the best one as we have confirmed by evaluating all possible alignments in this small example, where we can afford an exhaustive search.

- The NW alignment is over the entire length of two sequences:
  - the traceback starts from the lower right corner of the traceback matrix, and completes in the upper left cell of this matrix.

- The Needleman-Wunsch algorithm works in the same way regardless of the length or complexity of sequences and ***guarantees*** to find the best alignment.

- The Needleman-Wunsch algorithm is appropriate for finding the best alignment of two sequences which are
  - (i) of similar length;
  - (ii) similar across their entire lengths.

It will be Continued….